translations, rotations, smooth deformations (e.g., stretching or squeezing), and to noise. CNNs are only invariant or equivariant to translations but are not necessarily robust against rotations or smooth deformations.

In order to get the kind of invariance we want, we need a *relational model*—a model that is capable of identifying objects and relating them to one another. If we have an image of a cup on top of a table, and we train a CNN to identify the cup, it will perform well. But if we were to rotate the image 90 degrees, it would likely fail because it is not rotation-invariant, and our training data did not include rotated images. However, a (purely) relational model should, in principle, have no problem with this because it can learn to do relational reasoning. It can learn that "cups are on tables," and this relational description does not depend on a particular viewing angle. Hence, machine learning models with relational reasoning abilities can model powerful and generic relations between objects. *Attention models* are one way of achieving this and the topic of this chapter.

## 10.2. Relational reasoning with attention

There are many possible ways of implementing a relational model. We know what we want: a model that can learn how objects in input data are related to one another. We also want the model to learn higher-level features over such objects, just like a CNN does. We also want to maintain the composability of ordinary deep learning models so we can stack together multiple layers (such as CNN layers) to learn more and more abstract features. And perhaps most important of all, we need this to be computationally efficient so we can train this relational model on large amounts of data.

A generic model called *self-attention* meets all of these requirements, although it is less scalable than the other models we've looked at so far. Self-attention, as the name suggests, involves an attention mechanism in which the model can learn to attend to a subset of the input data. But before we get to self-attention, let's first talk about ordinary attention.

### 10.2.1. Attention models